Srikeerthi Srinivasan

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SUMMARY

Software Development Engineer with 2 years of production experience and a strong foundation in fault-tolerant architecture - comfortable owning systems end to end. Proficient in Python, JavaScript, SQL, and AWS. Published open-source contributor. MS Computer Science, UT Arlington.

SKILLS

- **Languages:** Python, JavaScript, Node.js, SQL, C#
- **Distributed Systems & Backend:** REST APIs, Event-Driven Architecture, Fault-Tolerant Design, Microservices, FastAPI, Django
- **AWS & Cloud:** Lambda, EC2, SQS, S3, API Gateway, CloudWatch, Secrets Manager, GCP, Docker, Kubernetes, CI/CD (GitHub Actions)
- **Data & Databases:** PostgreSQL, MongoDB, Kafka, Spark Streaming, Milvus, LakeFS, ETL Pipelines

EXPERIENCE

AI Engineer Intern | Olympic Collectibles AI Solutions, Renton, WA 12/2025 – 03/2026

- Diagnosed why a backend serving 150K+ requests stalled under heavy load - the GPU was being overwhelmed by too many simultaneous requests. Added concurrency controls that limit how many requests run at once, restoring stable throughput with no downtime.
- Built a large-scale data pipeline indexing 450+ GB of image data into Milvus with LakeFS versioning and SQL integrity validation - 99.15% retrieval precision with automated quality checks across every ingestion cycle.
- Designed a rigorous evaluation framework comparing two vision-language models (GLM-4.6V-Flash vs InternVL 2.5) across accuracy and latency - deployed the winning model at 98.6% accuracy with full output traceability.
- Diagnosed a classification accuracy bottleneck where single-crop inference missed fine-grained visual variants - redesigned the pipeline around DINOv2 multi-crop feature extraction, improving validation accuracy to 97.14%.

Software Developer Intern | ecohome.one LLC, Redmond, WA 01/2025 – 09/2025

- Built a voice-activated e-receipt parsing feature using Alexa Skills Kit and Gemini API with OAuth 2.0 - users speak a command, the system extracts purchase data from Gmail confirmation emails, and auto-populates the inventory app.
- Architected a fault-tolerant serverless backend on Node.js, AWS Lambda, and SQS with Dead Letter Queues - zero data loss across variable traffic loads with automatic recovery for failed events.
- Refactored a monolithic Laravel codebase using SOLID principles - decoupled database transactions from validation logic, achieving a 10% speed improvement and reducing production bug surface area.

Machine Learning Engineer | Scientist Technologies Pvt Ltd, Bengaluru, India 05/2021 – 07/2022

- Built a Mediapipe pose estimation system for physiotherapy patients - classifies exercise type, counts reps, and calculates joint angles and movement velocity at 90% accuracy, generating progress reports that therapists used to adjust treatment plans.
- Replaced an inefficient Tesseract OCR framework with a ResNet50 model for reading analog digits from medical devices - built automated Python validation scripts that caught a 9% accuracy drift in production, curated remediation data, and redeployed at 95% accuracy with no service disruption. Reduced processing cost by 5% by eliminating cloud OCR API calls.
- Built a Python + Scrapy data ingestion pipeline on AWS EC2 with anti-bot mitigation - mined and structured 5,000+ records into a clean production database for downstream model training.

PROJECTS

Real-Time Infrastructure Telemetry Pipeline | github.com/srikeerthis/Infrastructure-Telemetry-Pipeline

- A distributed observability platform on Kubernetes that ingests server telemetry in real time, persists it for historical analysis, and notifies on-call engineers when systems show signs of stress.
- Kafka + Spark Streaming with PySpark watermarking, 1-minute windowed aggregations into PostgreSQL StatefulSets, and a FastAPI metrics API.
- A standalone Kafka consumer flags CPU spikes above 80% via Grafana dashboards and Discord alerts.

SafeArc - Multi-Agent AI Safety Engine | github.com/srikeerthis/safearc

- A real-time vision system that watches a workspace through a camera, enforces geometric safety zones around humans, and generates safe action plans for robotic agents.
- Three-agent pipeline - Gemini API + OpenCV, custom Python orchestration - with shared state, explicit handoffs, deterministic safety enforcement, and fallback logic for unparseable outputs.
- Stored rated session history in SQLite and retrieved past scenarios by scene similarity - injecting them as few-shot context so the planner improves over time without retraining.

Real-Time Bidding Engine | github.com/srikeerthis/Real-Time-Bidding-Engine

- An online ad bidding engine using LinUCB contextual bandits - selects bid strategies per user context and learns from feedback continuously.
- Achieved 73.90% conversion rate in simulation at under 20ms latency - bid decisions return instantly while model updates happen in the background so they never slow down the live response.
- Stateless Dockerized microservice with decoupled inference and feedback paths for horizontal scaling.

OPEN SOURCE

cobot-safety | pypi.org/project/cobot-safety - Published Python library for geometric safety enforcement. Zero dependencies, Apache 2.0.

EDUCATION

MS Computer Science | University of Texas at Arlington | GPA: 3.46/4.00 07/2022 – 12/2024

BE Computer Science | Vidyavardhaka College of Engineering | GPA: 8.19/10.00 07/2017 – 08/2021